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General Knowledge
World Geography

STUDY NOTES

Climatology: Water in the Atmosphere

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The **significance of water in the atmosphere** could be understood by the following points:

- The amount of water vapour present in a given volume of air indicates the **atmosphere's potential capacity for precipitation**.
- Water vapour absorbs radiation and therefore it is a **regulator of heat loss** from the earth.
- The amount of water vapour decides the **quantity of latent energy** stored up in the atmosphere for the growth of storms.
- The quantity of water vapour in the atmosphere even affects the **human body's rate of cooling**.

Absolute humidity

- The weight of the **actual amount of water vapour** present in a unit volume of air is called the absolute humidity. It is usually expressed as grams per cubic metre of air.

Specific Humidity

- It is the weight of water vapour per unit weight of air or the **proportion of the mass of water vapour to the total mass of air** is called the specific humidity. It is measured in the units of weight (gram per kilogram).

Relative Humidity

- It is the ratio of the air's actual water vapour content to its water vapour capacity at a given temperature. It is expressed in percentage.
- With the change of air temperature, the capacity to retain moisture increases or decreases and the relative humidity is also affected. It is greater over the oceans and lesser over the continents.

Dew Point

- The temperature at which **saturation** occurs in a given sample of air or water vapour begins to change into water is known as dew point.

Evaporation

- It is the process by which water is transformed from liquid to gaseous form. It takes approximately 600 calories of energy to convert one gram of water to water-vapour.
- The energy absorbed by the water molecules during evaporation is used solely to give them the motion needed to escape the surface of the liquid and become a gas. Because this energy is subsequently released as heat when vapour changes back into liquid, it is generally referred to as **latent heat**.

The rate of evaporation depends on factors like:

- Temperature
- The moisture content of air
- Movement of air

If the moisture content is low, the air has a greater potential to absorb and retain a larger amount of moisture. Air with high humidity has less potential, and hence evaporation is slow and small.

Similarly, an increase in temperature increases the water absorption and retention capacity of air. Movement of air replaces the saturated layer with an unsaturated layer which has a greater capacity of absorbing moisture. Hence, the **greater the movement of air, the greater is the evaporation**.

Condensation

- It is the process of change of state from gaseous to liquid or solid state.
- When moist air is cooled, it may reach a level wherein its capacity to hold water vapour is exceeded by the actual amount present in it.
- The excess water vapour then condenses into a liquid or solid form depending upon the temperature. **It depends on: The amount of cooling and The relative humidity of the air.**
- Condensation in air itself can only take place if the temperature of the air is reduced to **below the dew point**.
- As the dew point of any mass of air is its saturation point, when its relative humidity is 100%, a little more cooling will bring about the point where condensation takes place, viz. water vapour changes into clouds or rain.
- In contrast, when relative humidity is low, a large amount of cooling is required to first reach the dew point and then condensation can take place.

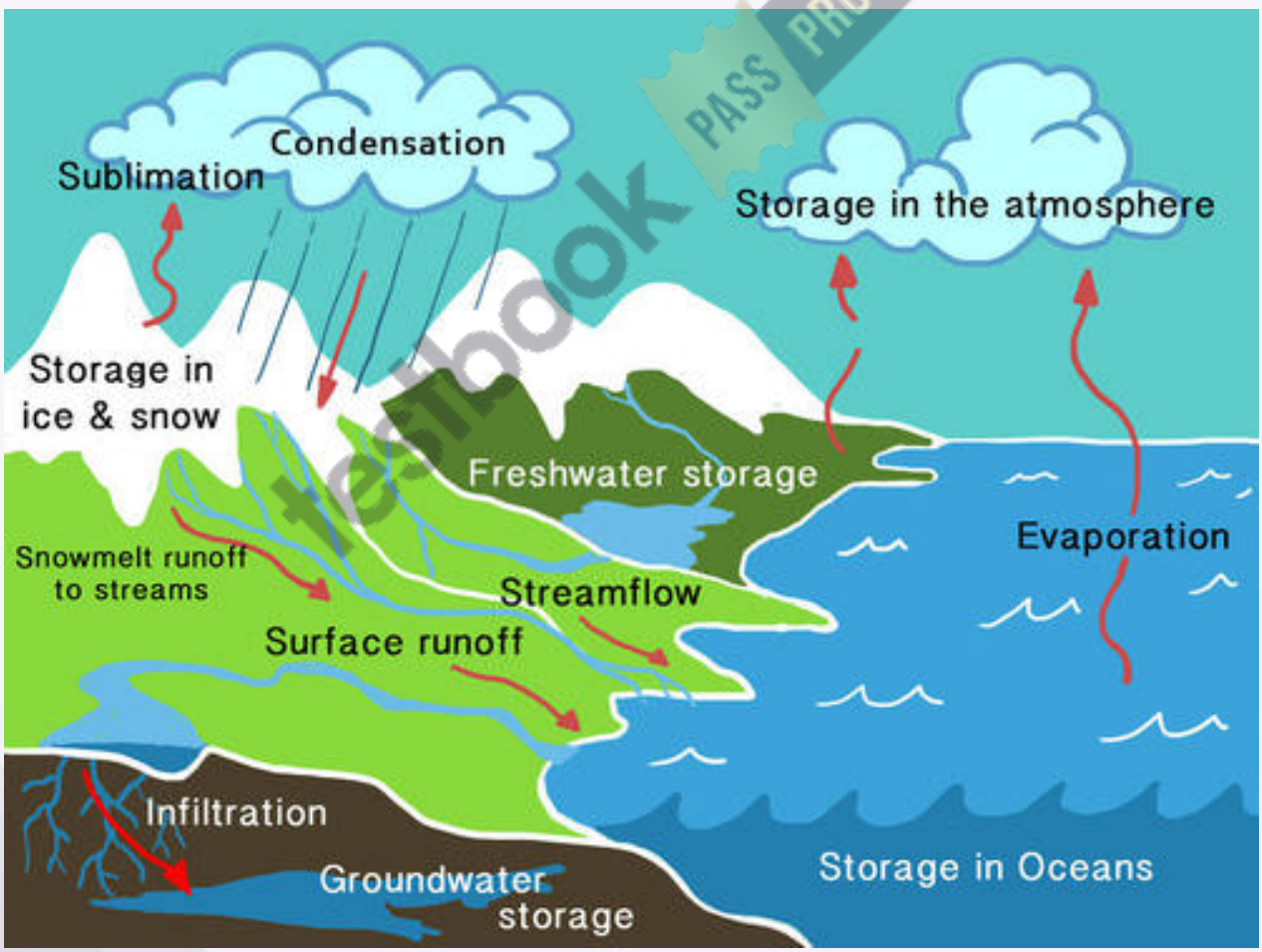
- If the moist air comes in contact with some colder object, the condensation may also take place when it is close to the dew point.

Forms of Condensation:

Forms of condensation can be classified on the basis of the temperature at which the dew point is reached. Condensation can take place when the dew point is-

- lower than the freezing point
- higher than the freezing point.

While frost, snow and some clouds are produced when the temperature is less than the freezing point, dew, fog and clouds result even when the temperature is higher than the freezing point.



Dew:

- When the moisture is deposited in the form of water droplets on cooler surfaces of stones, grass blades and plant leaves, it is known as *dew*.
- The ideal conditions for its formation are a clear sky, calm air, high relative humidity, and cold and long nights.
- For the formation of dew**, it is necessary that the dew point is above the freezing point.**

Frost:

- Frost forms on cold surfaces when condensation takes place below the freezing point (0°C), i.e. the **dew point is at or below the freezing point.**
- The excess moisture is deposited in the form of minute ice crystals instead of water droplets.
- The ideal conditions for the formation of white frost are the same as those for the formation of dew, except that the air temperature must be at or below the freezing point.

Fog and Mist:

- When the temperature of an air mass containing a large quantity of water vapour falls all of a sudden, condensation takes place within itself on fine dust particles.
- **Fog is a cloud with its base at or very near to the ground** and hence the visibility becomes poor to zero.
- In urban and industrial centres smoke provides plenty of nuclei which help the formation of fog and mist. Such a condition when fog is mixed with smoke is described as **smog**.
- The only difference between the mist and fog is that **mist contains more moisture than the fog**. In the mist, each nuclei contains a thicker layer of moisture.
- Mists are frequent over mountains as the rising warm air up the slopes meets a cold surface. Fogs are drier than mist and they are prevalent where warm currents of air come in contact with cold currents.